

Introduction to Action Recognition

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Input Source for Action Recognition

- Video
 - RGB image
 - Depth image
- Motion from Image
 - eg. optical flow
- Skeleton Data
 - Pose Estimation
- Sensor Data
 - eg. motion sensor

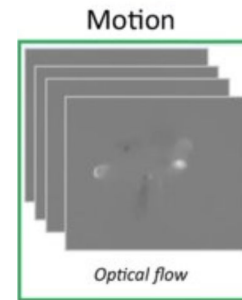


image: http://jd92.wang/assets/files/a10_pr18.pdf

Image architectures are wasteful for processing high-frame rate video

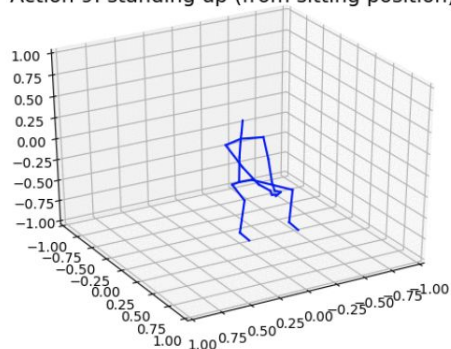


Human pose data?

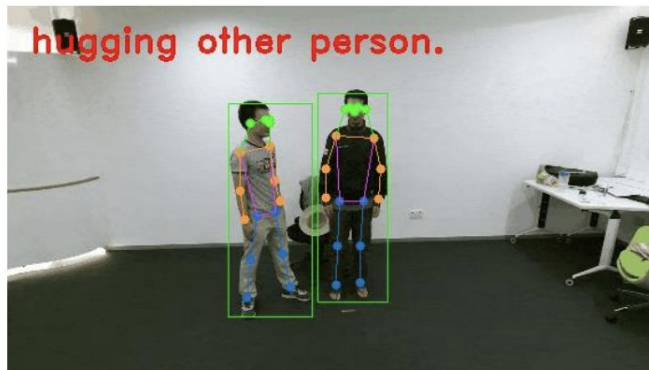


How to obtain human pose?

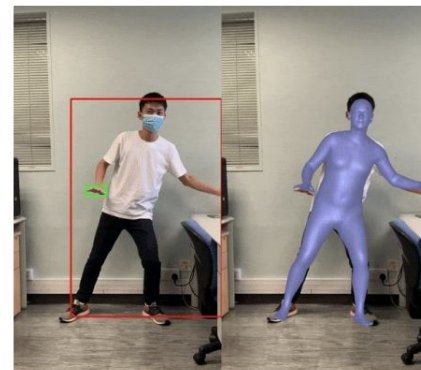
Action 9: standing up (from sitting position)



Kinect Sensor (RGBD)



Pose Estimation (2D / 3D)



Mocap (3D)

image: https://kennymckormick.github.io/files/cvpr22_tutorial.pdf

Pose Estimation

- Estimating the position of various parts of the human body



image: <https://supervisely.com/blog/human-pose-estimation/>

Pose Estimation Models

- OpenPose
- AlphaPose
- MediaPipe (Google)
- MoveNet (Google)
- YOLO (Ultralytics)
- etc.

YOLO Pose Estimation

- <https://docs.ultralytics.com/tasks/pose/>


ultralytics

Ultralytics YOLOv8 Docs

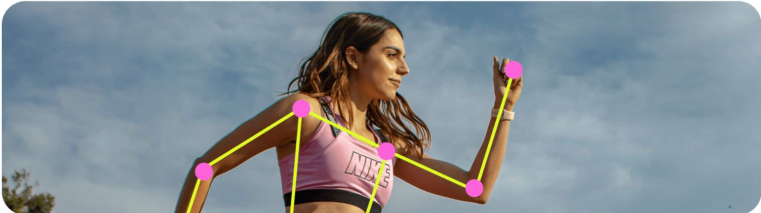


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Tasks

- Detect
- Segment
- Classify
- Pose**
- OBB

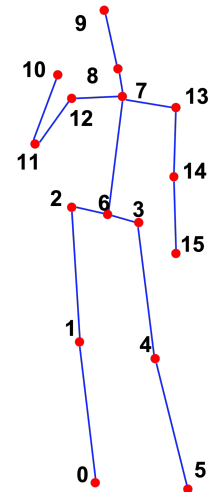
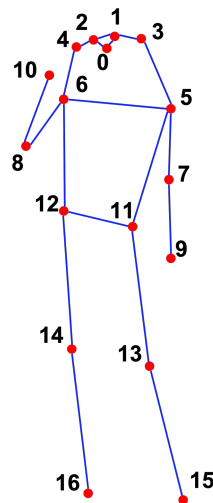
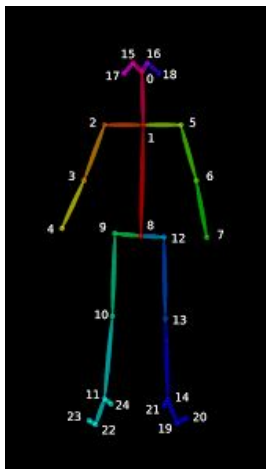
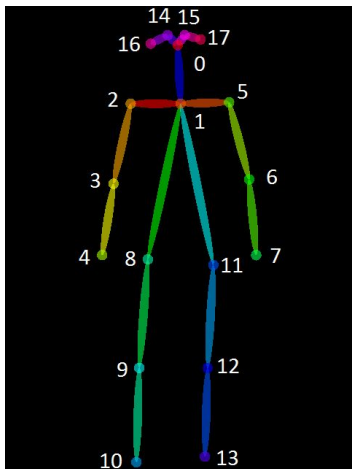
Pose Estimation



Pose estimation is a task that involves identifying the location of specific points in an image, usually referred to as keypoints. The keypoints can represent various parts of the object such as joints, landmarks, or other distinctive features. The locations of the keypoints are usually represented as a set of 2D $[x, y]$ or 3D $[x, y, \text{visible}]$ coordinates.

Keypoint Format

- Representation: Joints (nodes) and bones (edges)
- Different model may use different format
 - Different no. of body keypoints
 - Different connected body parts



How pose estimation is done?

- Bottom-up approach
 - model detects every instance of a particular keypoint (e.g. all left hands) in a given image
 - then attempts to assemble groups of keypoints into skeletons for distinct objects
- Top-down approach.
 - first uses an object detector to draw a box around each instance of an object
 - then estimates the keypoints within each cropped region.

How pose estimation is done?

Steps in OpenPose



(a) Input Image



(b) Part Confidence Maps



(c) Part Affinity Fields



(d) Bipartite Matching



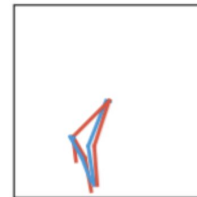
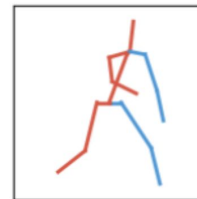
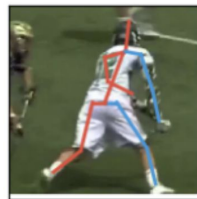
(e) Parsing Results

image: <https://kili-technology.com/data-labeling/machine-learning/human-pose-estimation-ultimate-beginners-guide-2023-edition>

2D vs 3D Human Pose Estimation

● 2D Human Pose Estimation

- Joint position is represented using X and Y coordinates
- pros:
 - easy to implement and computationally efficient
 - good choice for real-time pose estimation applications
- cons:
 - can only represent the human body in a 2D space
 - complex body positions will not be detected correctly

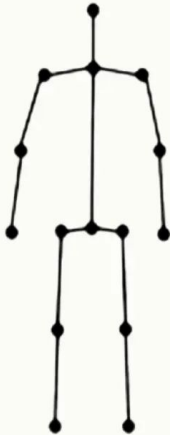


● 3D Human Pose Estimation

- Joint position is represented on the 3D space (X, Y, and Z)
- Transform a 2D image into a 3D object by estimating an additional Z-dimension to the prediction
- pros:
 - captures a large range of motion and can model more complex body positions
- cons:
 - time-consuming and requires more processing power than 2D pose estimation

Types of Human Pose Estimation Models

Skeleton-Based Model



Contour-Based Model



Volume-Based Model

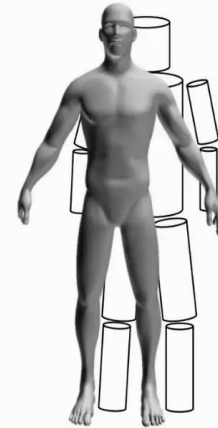


image: <https://kili-technology.com/data-labeling/machine-learning/human-pose-estimation-ultimate-beginners-guide-2023-edition>

Other Pose Estimations

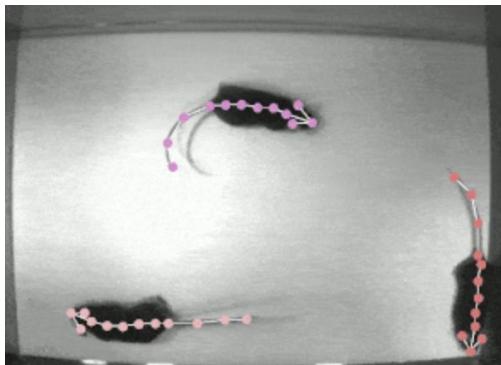
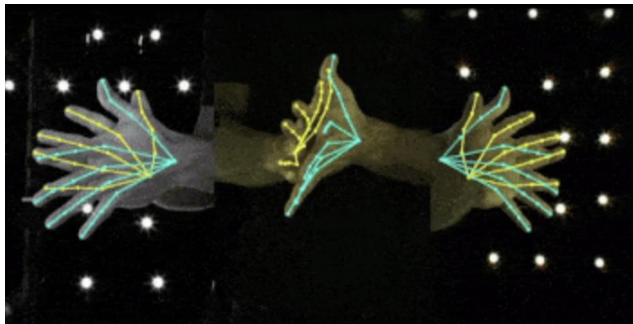
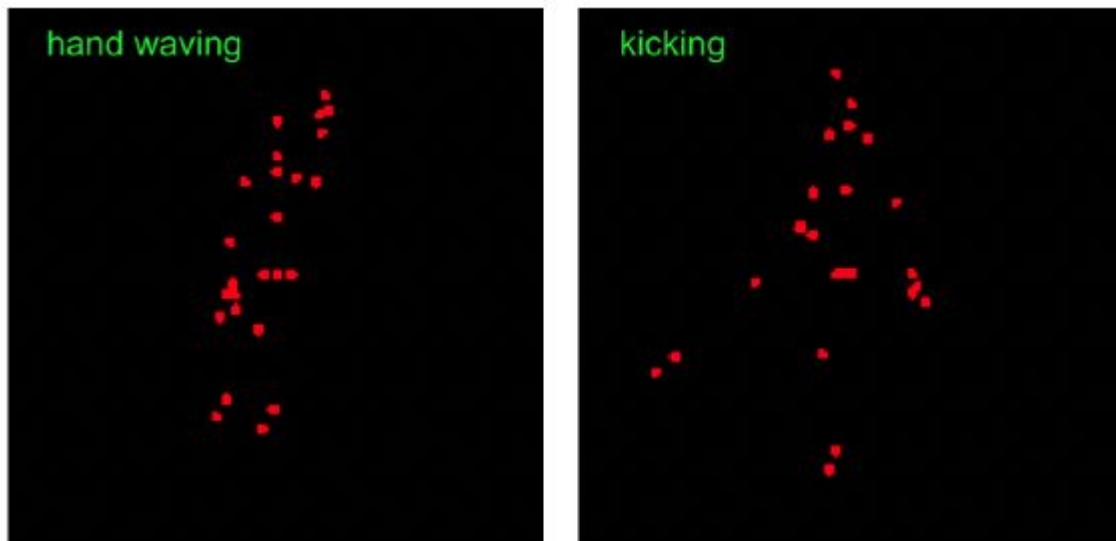


image: <https://kili-technology.com/data-labeling/machine-learning/human-pose-estimation-ultimate-beginners-guide-2023-edition>

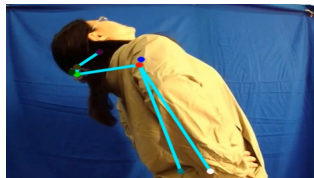
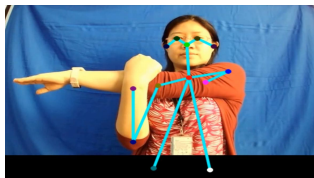
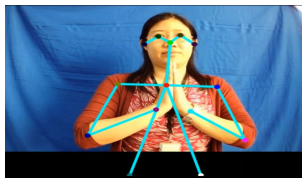
Skeleton-based Action Recognition



Challenges in Skeleton-based Action Recognition

- Quality of skeleton data

- missing skeleton data
- incorrect skeleton data



- For no-hand pose structure, low performance of action related to hands

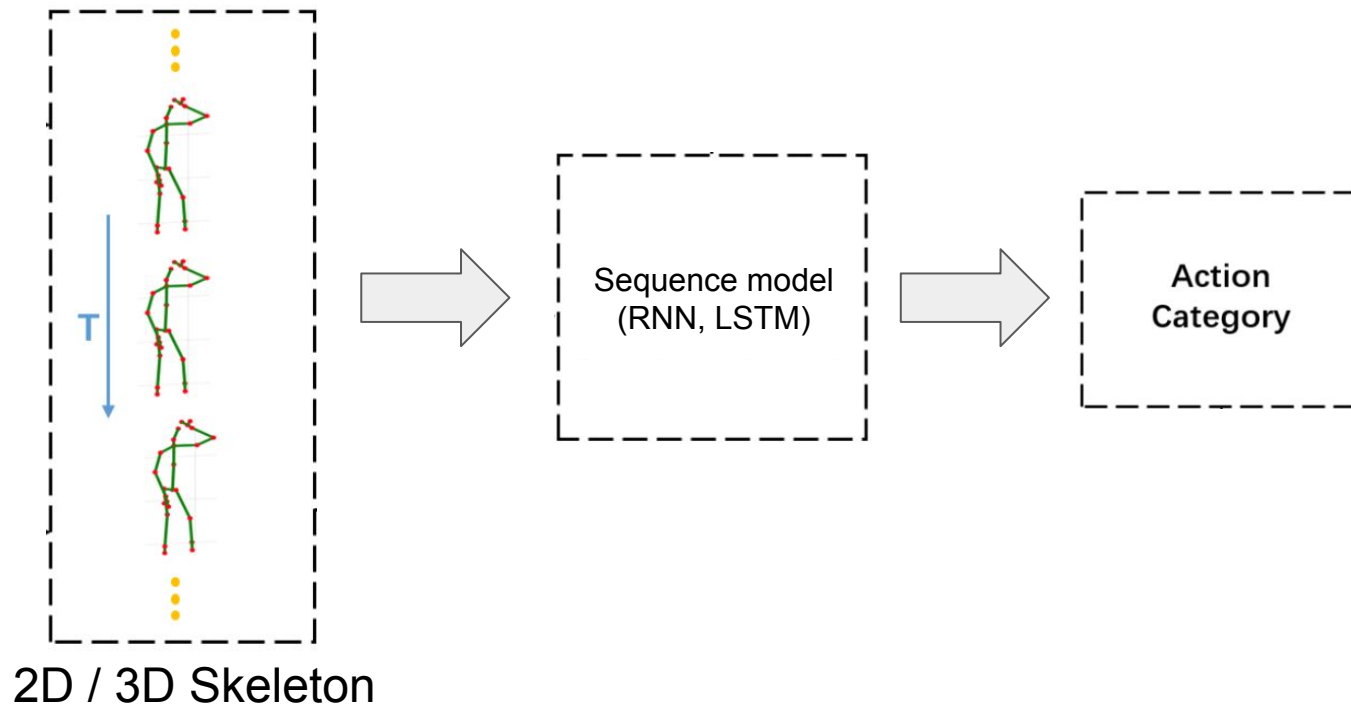
73: staple book
72: make victory sign
71: make OK sign
012: writing
76: cutting paper
011: reading
75: cutting nails
103: yawn
106: hit with object
74: counting money

Top10 worst performance on NTU-RGBD120

Skeleton-based Action Recognition Models

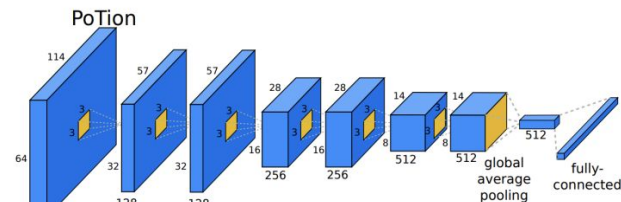
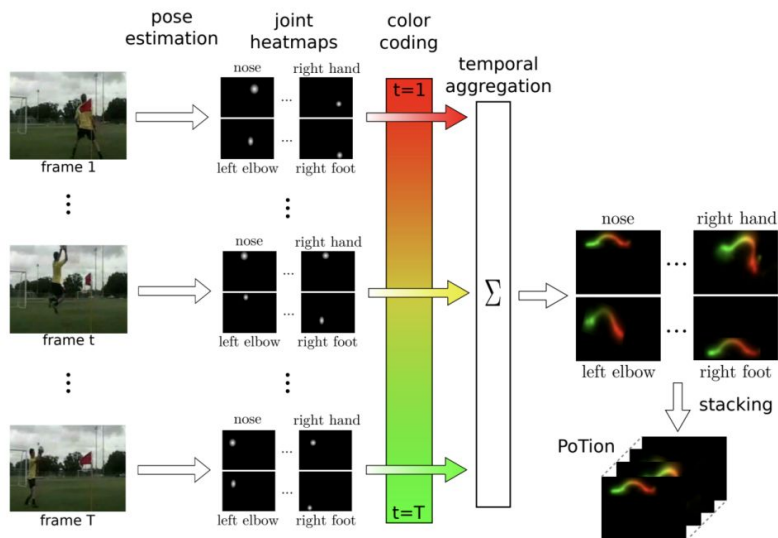
- Sequence model
- 2D CNN based model
 - PoTion
- 3D CNN based model
 - PoseConv3D
- Graph convolution network
 - ST-GCN

Sequence Model



PoTion

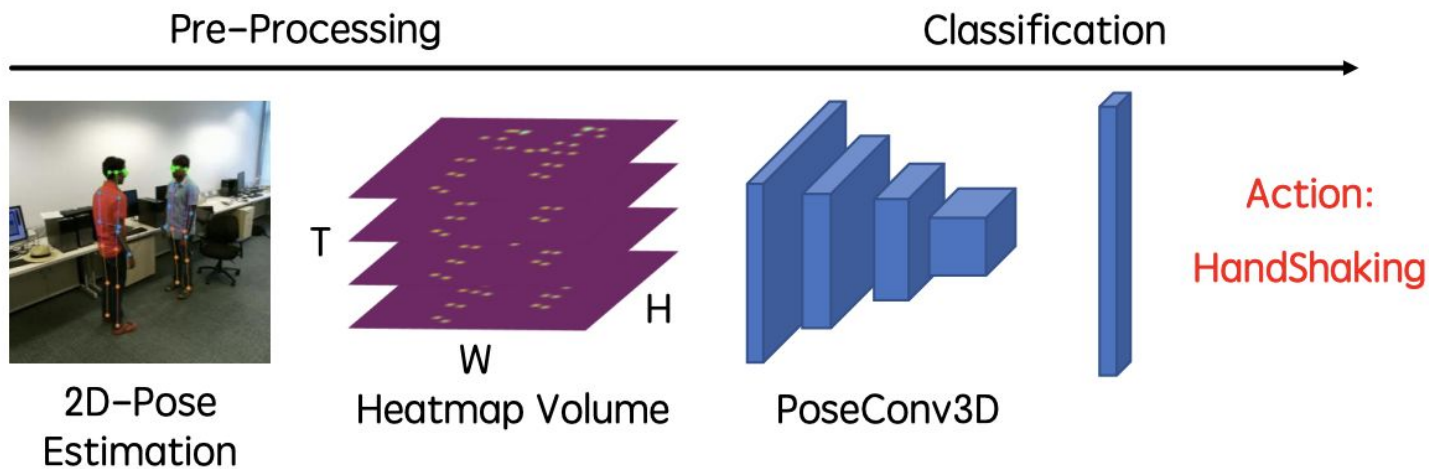
- Use 2D CNN
- Input: pseudo image



Choutas et al., Potion: Pose motion representation for action recognition, CVPR 2018

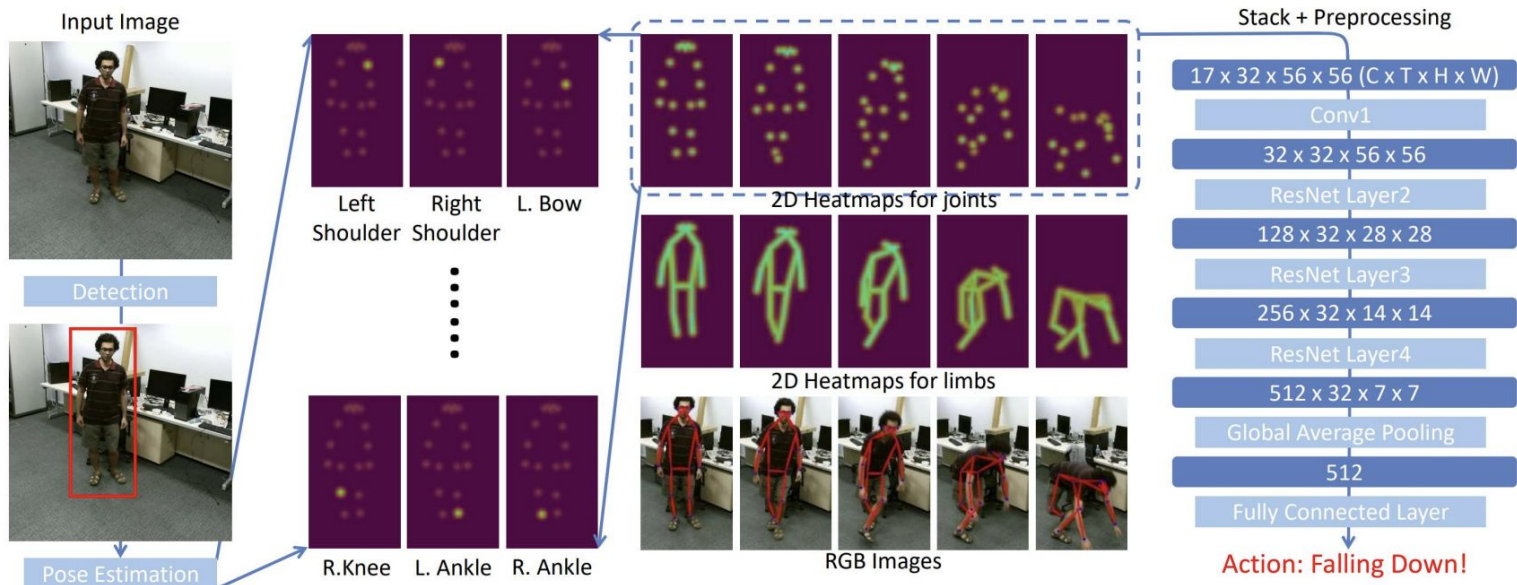
PoseConv3D

- Use 3D CNN
- Input: heatmap volumes



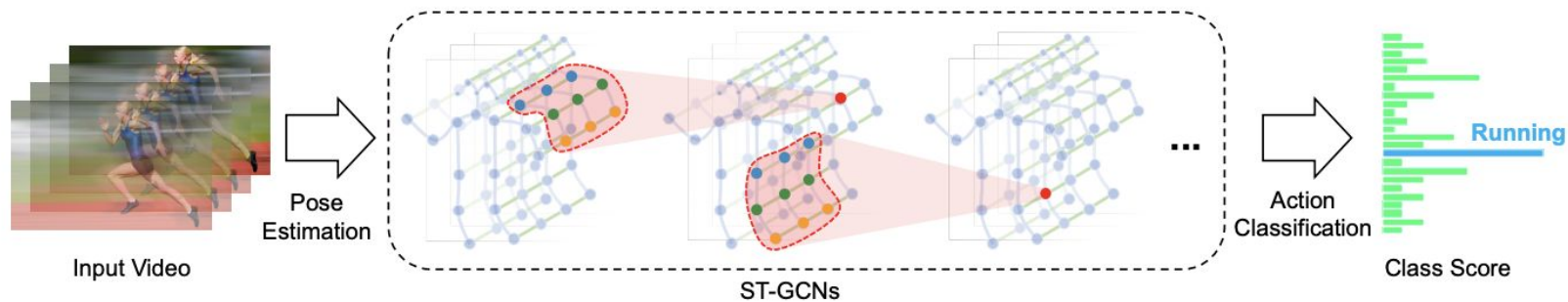
Duan et al., Revisiting skeleton-based action recognition, CVPR 2018

PoseConv3D



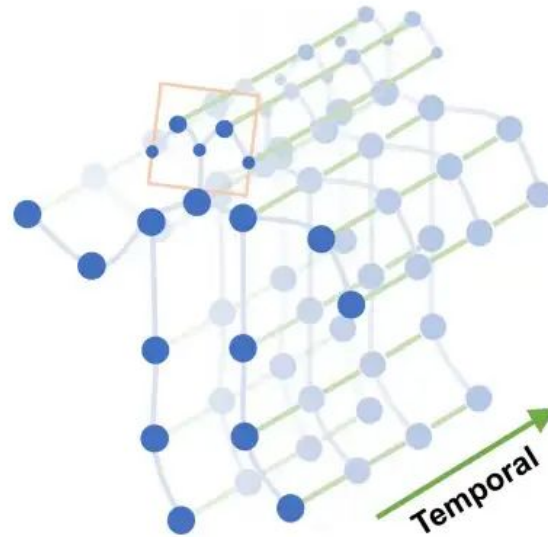
Duan et al., Revisiting skeleton-based action recognition, CVPR 2018

ST-GCN

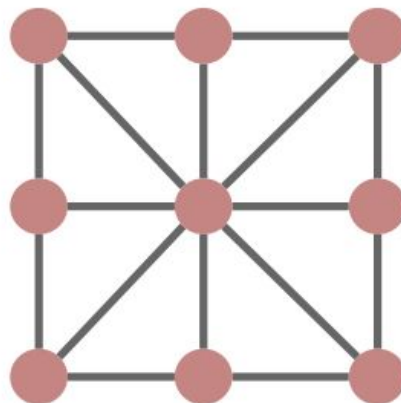
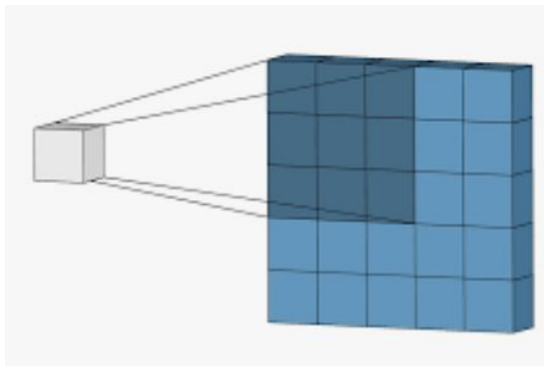


Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition, AAAI 2018

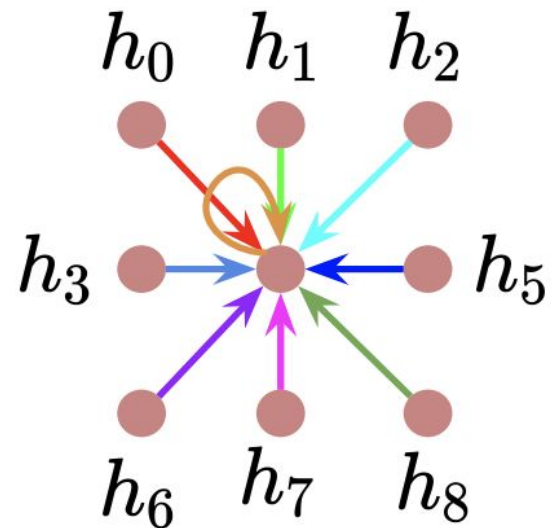
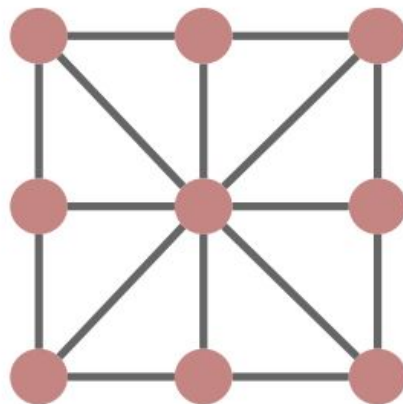
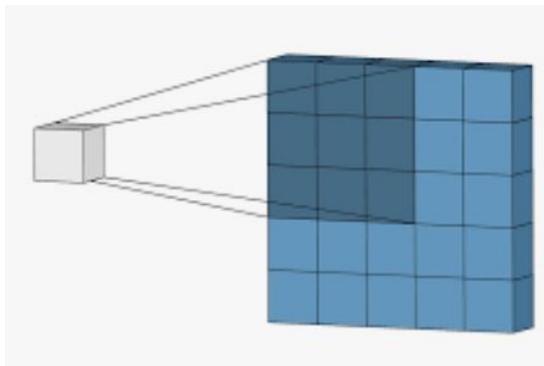
Graph Convolution Network



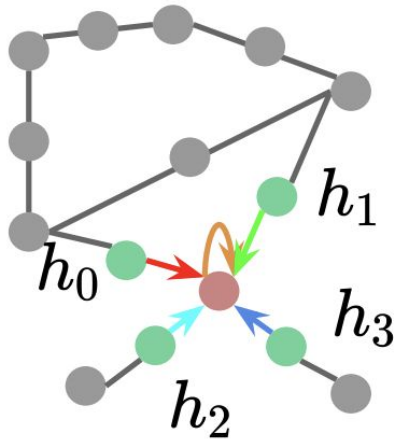
Graph Convolution



Graph Convolution



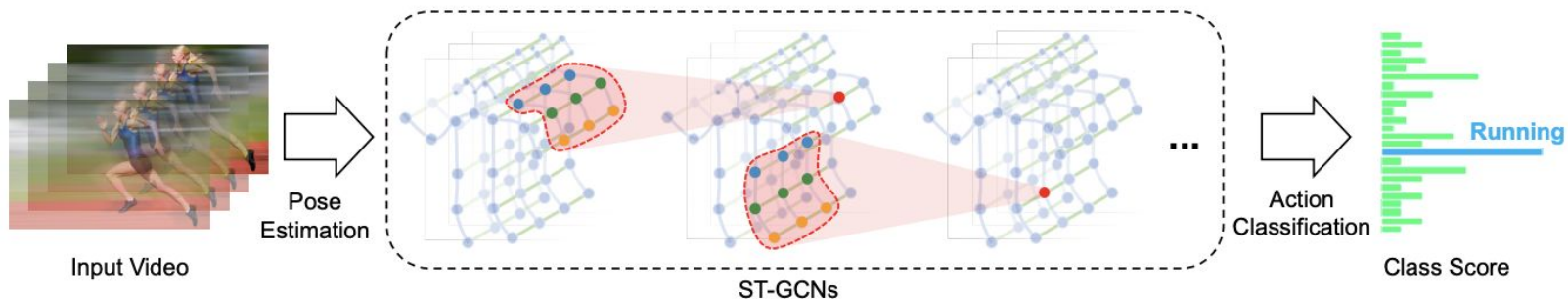
Graph Convolution



$$h_i^{l+1} = W h_i^l + \sum_{j \in N(i)} \frac{1}{c_{ij}} W h_j^l$$

ST-GCN

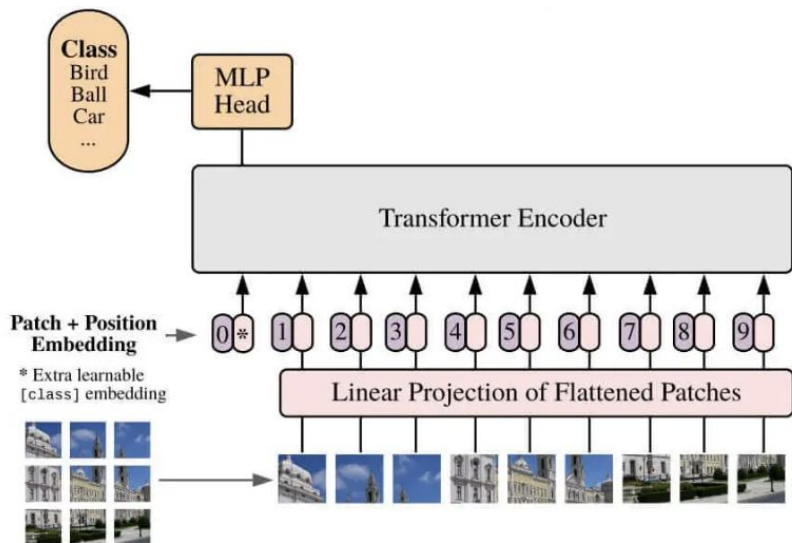
- Graph convolution network
- Convolutions over neighboring nodes collected in both spatial and temporal directions



Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition, AAAI 2018

Transformer-based Model

- Vision transformer model



Video Vision Transformer

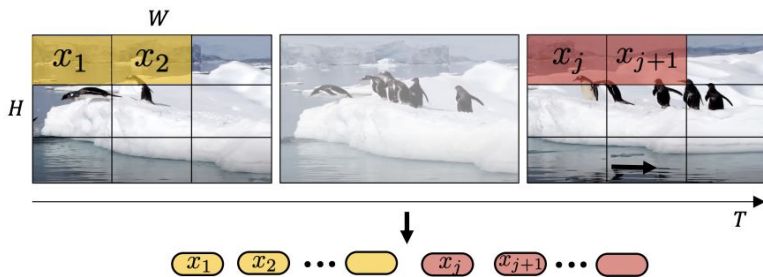


Figure 2: Uniform frame sampling: We simply sample n_t frames, and embed each 2D frame independently following ViT [17].

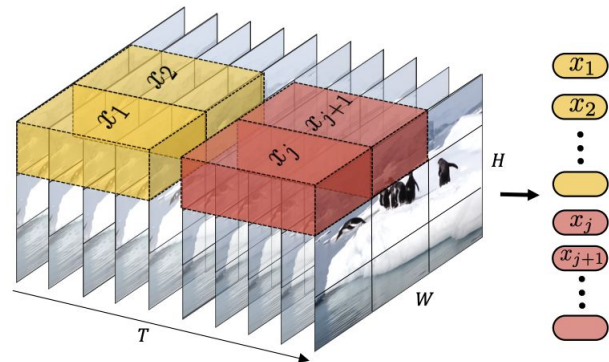
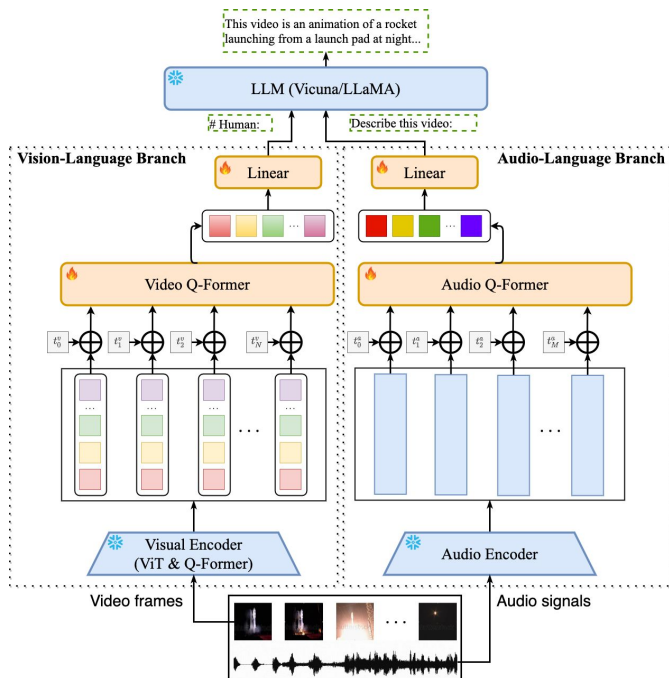


Figure 3: Tubelet embedding. We extract and linearly embed non-overlapping tubelets that span the spatio-temporal input volume.

Video-LLaMA

- An Instruction-tuned Audio-Visual Language Model for Video Understanding



- Video Q-Former (Query Transformer)
 - Learnable query tokens that attend to visual (or video) features.
 - A transformer architecture that outputs compact, semantic representations of the input.
 - high-dimensional visual input -> a small set of language-aligned embeddings

image: <https://github.com/DAMO-NLP-SG/Video-LLaMA>



How was the audience's response?



The audience was really **enthusiastic** and **clapped loudly** for the **performer**.

What is this man doing?



He is playing a **saxophone** on stage at the **concert**.



Summarize this video in one sentence.



The video shows a beautiful scenery of a **cherry blossom-lined** river **flowing by a boat** on the water, and a cityscape with **tall buildings** in the background.

What direction is the ship going



The ship is going towards the **right** side of the video.

Action Datasets

- Many action datasets available for research

Dataset	RGB	S	D	IR	PC	ES	Au	Ac	Gyr
UAV-Human [275]	✓	✓	✓	✓	✓	✓	x	x	x
HMDB51 [189]	✓	x	x	x	x	x	x	x	x
Kinectis-400 [190]	✓	x	x	x	x	x	x	x	x
Kinectis-600 [276]	✓	x	x	x	x	x	x	x	x
Kinectis-700 [277]	✓	x	x	x	x	x	x	x	x
EPICKITCHENS-55 [278]	✓	x	x	x	x	x	x	x	x
UCF-101 [188]	✓	x	x	x	x	x	x	x	x
HMDB-51 [189]	✓	✓	✓	✓	x	x	x	x	x
EPICKITCHENS-55 [278]	✓	x	x	x	x	x	✓	x	x
THUMOS Challenge 15 [279]	✓	x	x	x	x	x	x	x	x
ActivityNet [280]	✓	x	x	x	x	x	x	x	x
Something-Something-v1 [281]	✓	x	x	x	x	x	x	x	x
Something-Something-v2 [281]	✓	x	x	x	x	x	x	x	x
NTU RGB+D [191]	✓	✓	✓	✓	x	x	x	x	x
NTU RGB+D 120 [192]	✓	✓	✓	✓	x	x	x	x	x
MSRDailyActivity3D [282]	✓	✓	✓	x	x	x	x	x	x
Northwestern-UCLA [283]	✓	✓	✓	x	x	x	x	x	x
UWA3D Multiview [284]	✓	✓	✓	x	x	x	x	x	x
UWA3D Multiview II [285]	✓	✓	✓	x	x	x	x	x	x
InfAR [241]	x	x	x	✓	x	x	x	x	x
DysGesture [286]	x	x	x	x	x	✓	x	x	x
DHP19 [243]	x	✓	x	x	x	✓	x	x	x
MMAAct [287]	✓	✓	x	x	x	x	x	✓	✓
UTD-MHAD [288]	✓	✓	x	x	x	x	x	✓	✓
PKU-MMD [254]	✓	✓	✓	✓	x	x	x	x	x
UCFKinect [289]	x	✓	x	x	x	x	x	x	x
HAA500 [290]	✓	x	x	x	x	x	x	x	x
UAV-Humman [275]	✓	✓	✓	✓	x	x	x	x	x
NEU-UB [291]	✓	x	✓	x	x	x	x	x	x
AVA [292]	✓	x	✓	x	x	x	x	x	x

S: Skeleton, D: Depth, IR: Infrared, PC: Point Cloud, ES: Event Stream, Au: Audio, Ac: Acceleration, Gyr: Gyroscope, EMG: Electromyography informations

Action Datasets

RGB image:

- ActivityNet
- UCF101 / THUMOS
- Kinetics 400/600/700



Skeleton data:

- NTU-RGB+D
- Kinetics-Skeleton
- Northwestern-UCLA Multiview Action 3D
- UTKinect-Action3D



image: <https://arxiv.org/pdf/2012.06567.pdf>

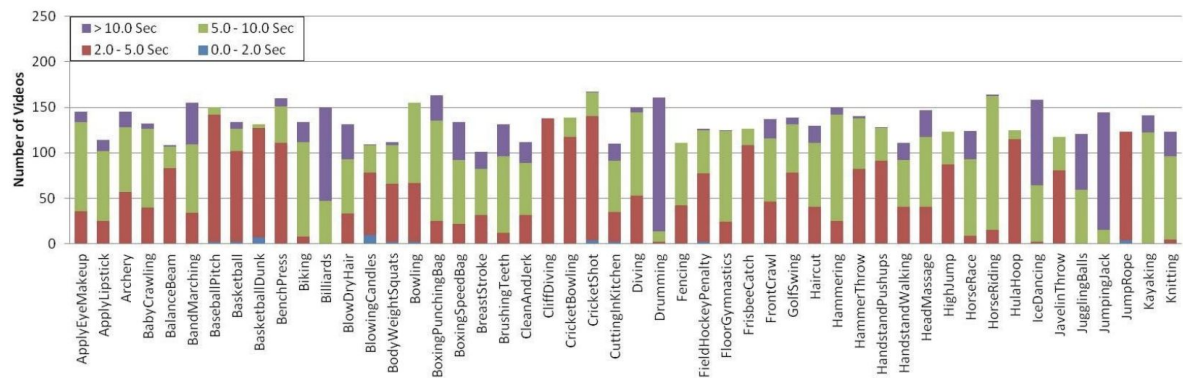
ActivityNet 1.3

- YouTube videos
- 10,024 videos
- 200 actions
- untrimmed
- Large variations in camera motion, object appearance and pose, viewpoint, background, illumination, etc.



UCF101 (THUMOS)

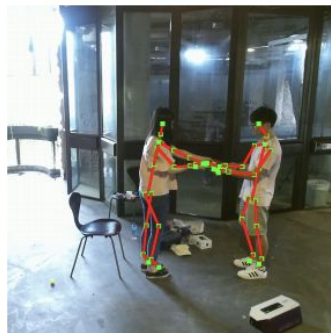
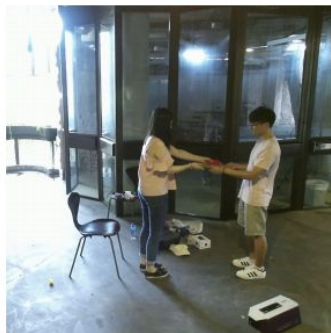
- YouTube videos
- 13,320 videos
- 101 actions
- trimmed



images: <https://www.crcv.ucf.edu/data/UCF101.php>

NTU-RGB+D Dataset

- 60 classes (v1), 120 classes (v2)
- 114,480 videos
- RGB, 3D Skeleton, Depth
- trimmed



Action Datasets

Other domains:

- First person
 - Epic-Kitchen
 - <https://epic-kitchens.github.io/2025>
- Instructional video
 - COIN (Comprehensive Instructional Video Analysis)
 - <https://coin-dataset.github.io/>
- Hand movement
 - Something-something
 - <https://www.qualcomm.com/developer/software/somethin-g-something-v-2-dataset>



make glutinous rice dumpling



install the bicycle rack



image:

Various Types of Actions

- Single-person action
- Human-human interaction
- Human-object interaction
- Group activity

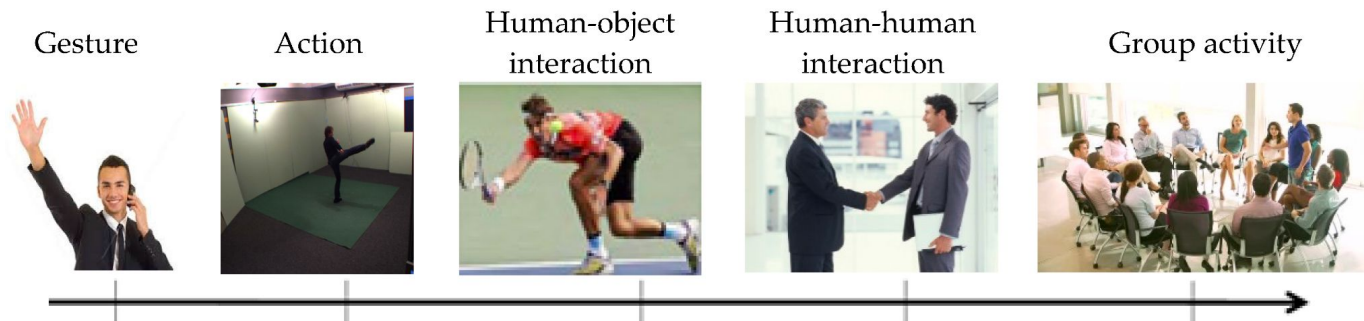


image from "A Comprehensive Review on Handcrafted and Learning-Based Action Representation Approaches for Human Activity Recognition", Appl. Sci., 2017

Challenges in Action Recognition

- Viewpoints

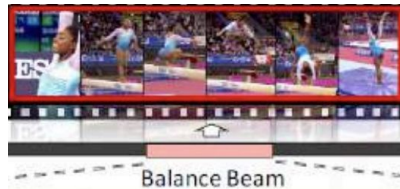


- Action similarities and variations

- eg. walking vs running, kicking -> various ways of kicking

Challenges in Action Recognition

- Different action length
 - Some actions are short, while some actions are long



- Occlusion

Challenges in Action Recognition

- Annotation is subjective
 - In some actions, start and end time are not easily defined eg. drinking
- Multiple actions in a video
- Multiple people in a video

Key Takeaways

- Action recognition in the wild is still a long way to go
- Many researches focus on single-person action
 - Real life videos contain multiple persons and not just single-person actions
- Realtime deployment for video analytics is challenging

Q&A